# THE INDIAN FATTY LIVER PLAYBOOK 2026-27

Read your report. Understand your numbers.
Reverse the fat before it becomes scarring.

Nearly four in ten Indian adults are walking around with fat in the liver and no symptoms at all. Most were told to 'just lose weight' and sent home. This book explains the report, the numbers, the food, the tests, the medicines approved in India, and the twelve weeks that actually change the scan - with nothing rounded up to sound scarier than it is.

**25 CHAPTERS | 8 PRINTABLE WORKSHEETS | SOURCES LISTED IN FULL**

Built from ICMR-NIN Dietary Guidelines 2024, INASL guidance, the MoHFW NP-NCD operational guidelines, EASL-EASD-EASO 2024, AASLD guidance and CDSCO approvals current to November 2026.

**DIGITAL EDITION**

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# How to use this book

Three sittings, one page of notes, and a plan you can actually keep.

This book exists because of a sentence that gets said in Indian clinics every day: "There is some fat in your liver. Lose weight and come back."

It is not wrong advice. It is just not enough. Between that sentence and the actual work of getting the fat out of your liver there are about fifteen questions: Which grade is serious? Is my ALT dangerous? Do I need a FibroScan? Is saroglitazar worth it? Can I still eat rice? What do I do about the Liv.52 my uncle swears by? Do I need to worry about cirrhosis? This book answers those questions in the order you will actually ask them.

It is written for the person holding the report, not for a doctor. Everything in it is drawn from Indian and international guidance that you can check yourself - the ICMR-NIN Dietary Guidelines for Indians 2024, the Indian National Association for Study of the Liver (INASL) guidance, the Ministry of Health's operational guidelines for fatty liver under the national NCD programme, the 2024 EASL-EASD-EASO clinical practice guidelines, AASLD guidance, and the Indian drug regulator's own approval record. Every number in this book has a source listed at the back. Nothing here is invented, and nothing is rounded up to sound scarier than it is.

## Read it in this order

1. **Sitting one (about 40 minutes):** Chapters 1 to 4. This is the part that turns a confusing report into a clear picture. By the end you will know whether you have fat alone or fat plus scarring, and why that distinction is the only one that really matters.
2. **Sitting two (about 30 minutes):** Chapters 5 to 9. Sit with your actual reports and a calculator. You will compute your own FIB-4 score and read your own LFT. Keep a pen handy - the worksheets in Part Six are meant to be filled in, not admired.
3. **Sitting three (about 40 minutes):** Chapters 10 to 16. This is the plan itself: your weight target, your plate, your walking, your twelve weeks. Then read Part Four before you buy a single supplement.

Parts Five and Six are reference material. Read them when the situation arrives - the monitoring calendar when your next test is due, the red-flag chapter if something changes quickly, the money chapter when you are staring at a bill.

**What this book is not**

It is not a diagnosis and it is not a substitute for your doctor. Fatty liver is a condition where the decisions - whether to start a medicine, whether to refer you to a hepatologist, whether a scan result is believable - depend on things a book cannot see. What this book does is make you the best-informed person in that consulting room. It tells you which questions to ask, which tests are worth paying for, and which claims are being sold to you.

**A note on dates**

Every fact here was checked against primary sources in November 2026. Fatty liver is a fast-moving field: the name changed in 2023, the first MASH drug was approved in the United States in 2024, and India approved semaglutide for MASH in July 2026. Where guidance is likely to shift within a year, this book says so instead of pretending otherwise.

CONTENTS

# Contents


| | | |
|---|---|---|
| | How to use this book | 2 |
| | **WHAT THE REPORT ACTUALLY SAYS** | |
| 1 | Fatty liver on your ultrasound: what grade 1, 2 and 3 really mean | 4 |
| 2 | The 2026 names: NAFLD, MASLD, MASH, MetALD - and why they changed | 5 |
| 3 | Why this is an Indian problem: the numbers, the thin-fat body, your waist | 7 |
| 4 | The number that predicts your future: fibrosis, not fat | 8 |
| | **READING YOUR OWN NUMBERS** | |
| 5 | Your LFT, line by line - and why a normal ALT proves nothing | 10 |
| 6 | FIB-4 in four minutes: the free score that decides your next step | 11 |
| 7 | FibroScan: CAP, kPa, cost, and how to prepare so the reading is real | 12 |
| 8 | What else it could be: the tests that rule out the other causes | 13 |
| 9 | Build your own risk map | 14 |
| | **THE TWELVE-WEEK REVERSAL PLAN** | |
| 10 | Your weight target: the 3 per cent, 7 per cent and 10 per cent rules | 16 |
| 11 | The Indian thali, rebuilt: what ICMR-NIN 2024 actually recommends | 17 |
| 12 | The five things to cut first, and the ones not worth bothering with | 18 |
| 13 | Six breakfasts, ten lunches and dinners, one seven-day menu | 19 |
| 14 | Movement: 200 minutes a week, and the walk after dinner | 21 |
| 15 | Sleep, stress, shifts, and the apnoea you may not know you have | 22 |
| 16 | The twelve-week tracker | 22 |
| | **MEDICINES, SUPPLEMENTS AND TRAPS** | |
| 17 | What is actually approved in India in 2026 | 24 |
| 18 | The supplement trap: liver tonics, detox kits and the DILI numbers | 25 |
| 19 | The medicines you already take: statins, paracetamol, TB drugs, thyroid | 26 |
| 20 | Vaccines, alcohol, and the "just one peg" question | 27 |
| | **LIVING WITH IT** | |
| 21 | Your monitoring calendar for the next five years | 29 |
| 22 | Red flags: when to stop reading and go to a hospital | 29 |
| 23 | Your family, your children, your kitchen | 30 |
| 24 | Money: tests, insurance, government schemes, and what a transplant costs | 31 |
| 25 | The honest twelve-month review: what "reversed" actually means | 32 |
| | **WORKSHEETS** | |
| W1-W8 | Eight printable worksheets | 34 |
| | Sources: where every number in this book comes from | -- |

PART ONE

# What the report actually says

Four chapters to turn a one-line ultrasound impression into a picture you can act on.

CHAPTER 1

## Fatty liver on your ultrasound: what grade 1, 2 and 3 really mean

The scan report usually says something like: *"Liver shows increased echogenicity. Grade I fatty change. Rest of the abdomen normal."* And that is all you get. No explanation, no number to track, no idea whether to worry. So let us take the sentence apart.

An ultrasound machine does not see fat. It sees **brightness**. Sound waves bounce back more strongly from tissue that contains fat droplets, so a fatty liver looks brighter than the kidney sitting next to it. The sonographer compares the two and writes down what they see. That comparison - not any measurement - is what the word "fatty" on your report is based on.

Indian labs almost always grade it in three lines. The grading is a rough visual impression, and different sonographers will grade the same liver differently. That is worth knowing before you build your life around the word "grade".

| What the report says | What the sonographer saw | What it means in practice |
|---|---|---|
| **Grade 1 (mild)** | Liver slightly brighter than normal; the blood vessels inside it are still clearly visible. | Fat is present, usually under about a third of liver cells. Almost never scarring at this stage - but the metabolic problem that caused it is already running. |
| **Grade 2 (moderate)** | Brightness clearly increased; vessels are partly obscured; the diaphragm may be dimmer. | More fat, and a higher chance that inflammation has started. This is the grade where most people are told to lose weight and most people do not, because they feel fine. |
| **Grade 3 (severe)** | Marked brightness; vessels and the deep part of the liver are hard to see. | A lot of fat. Still not the same as scarring, but this is where the odds of fibrosis climb - and where a FibroScan stops being optional. |

Two things about this table matter more than the grades themselves.

**First, the grade does not tell you about scarring.** Ultrasound is good at fat and poor at fibrosis. Published estimates put its sensitivity for detecting a fatty liver at around 85 per cent with specificity around 95 per cent - respectable for fat - but it drops in people with obesity, and it simply cannot stage scarring. A person with grade 1 fat and fibrosis stage 2 is in more danger than a person with grade 3 fat and no fibrosis. The grade you were given measures the least important of the two things in your liver.

**Second, "grade 1" is not a mild disease.** It is a mild *appearance*. The metabolic machinery that put the fat there - insulin resistance, a liver making fat out of sugar, visceral fat pressing on the system - does not care what grade the sonographer wrote. Roughly four in ten Indian adults have some degree of fatty liver, and most of them have never been told anything beyond the grade.

> **The first week, in five steps**

1. Find the actual report, not the WhatsApp photo of it. You need the LFT values (ALT, AST, GGT, platelets) that came with the scan. If you never had blood tests, that is the first thing to fix - a scan alone cannot be acted on.
2. Note your weight, your waist measured at the navel, and your blood pressure. Write them down with the date.
3. Read Chapter 6 and compute your FIB-4 score. It takes four minutes and costs nothing.
4. If your FIB-4 is above 1.3, book a FibroScan. If it is below 1.0, you have time - use it.
5. Do not buy a liver tonic. Chapter 18 explains why in detail.

One more thing worth saying plainly, because it is the most common misunderstanding in the waiting room: fatty liver is not caused by eating ghee, and it is not caused by "heat" in the body, and it is not a liver that needs cleaning. It is a metabolic condition in which the liver accumulates fat because the body is handling energy badly. That is why the treatment is food, movement, sleep and - in some people - medicine, and why no detox product has ever been shown to change it.

CHAPTER 2

## The 2026 names: NAFLD, MASLD, MASH, MetALD - and why they changed

If you have been reading about this condition online you will have met three or four different names for what may be the same thing in your body. That is not you misreading. The naming changed, and Indian reports, labs and doctors are still catching up.

| Name you will see | What it means | Status |
|---|---|---|
| **NAFLD (non-alcoholic fatty liver disease)** | Fat in the liver, not caused by alcohol. The old name, used in most Indian reports and in the 2023 INASL guidance. | Still widely printed. Now called MASLD. |
| **MAFLD** | Metabolic dysfunction-associated fatty liver disease - a 2020 proposal that made metabolic dysfunction a positive criterion. Used in some research. | Superseded by MASLD. |
| **MASLD (metabolic dysfunction-associated steatotic liver disease)** | Fat in the liver plus at least one cardiometabolic risk factor, with alcohol below the harmful threshold. The current international name; adopted by INASL in 2025. | Current. Use this one. |
| **MASH (metabolic dysfunction-associated steatohepatitis)** | MASLD plus inflammation and cell injury - the form that scars. Previously called NASH. | Current. |
| **MetALD** | MASLD plus alcohol intake above the MASLD threshold but below clearly harmful levels. | Current. |
| **ALD (alcohol-related liver disease)** | Alcohol is the dominant driver. | Current. |

The reason for the change is not bureaucratic fashion. The old definition was a definition of *exclusion*: you had NAFLD if you had fat and *no other cause* - no alcohol, no hepatitis, no drugs. That meant a person with fatty liver who also drank, or who also had hepatitis B, fell into a diagnostic no-man's land. The new definition is a definition of *inclusion*: if you have fat and at least one metabolic risk factor, you have MASLD, whatever else is also going on. It also removed the word "alcoholic" from the label of a disease that most people with it never caused with alcohol, and that stigma was doing real harm - people were not going to follow-up because of what they thought the report said about them.

For you, practically, the change matters in two ways. It means your report saying NAFLD and a newer article saying MASLD are talking about the same liver. And it means you should be checked for the other

causes *alongside* the metabolic one - Chapter 8 lists exactly which tests do that. Studies of Indian cohorts have found that 89 to 99 per cent of people who met the old NAFLD criteria also meet the new MASLD criteria, so the population is essentially the same people.

## The five criteria - and how to tell whether you meet them

MASLD requires fat in the liver plus at least one of five cardiometabolic criteria. Look at your own reports and tick what applies:

1. **Weight or waist:** BMI of 25 or more (23 or more if you use the Asian Indian cut-offs your doctor should be using), or a waist above 94 cm in men and 80 cm in women.
2. **Blood sugar:** fasting glucose of 100 mg/dL or more, or HbA1c of 5.7 per cent or more, or a diagnosis of type 2 diabetes, or you are on glucose-lowering medicine.
3. **Blood pressure:** 130/85 mmHg or higher, or you take a blood-pressure medicine.
4. **Triglycerides:** 150 mg/dL or more, or you take a lipid-lowering medicine.
5. **HDL cholesterol:** 40 mg/dL or less in men, 50 mg/dL or less in women, or you take a lipid-lowering medicine.

Almost everyone with fatty liver in India meets at least one of these, because the fat and the metabolic problem are the same problem wearing two hats. If you meet none of them and you have fat in the liver, that is unusual enough to be worth investigating properly - Chapter 8.

## Where alcohol fits, in grams rather than in pegs

Indian guidance uses daily alcohol in grams, and the thresholds are specific: below 20 g a day for women and below 30 g a day for men counts as the MASLD range; 20 to 50 g a day in women and 30 to 60 g a day in men puts you in MetALD, where both alcohol and metabolic dysfunction are doing damage; at 50 g a day in women or 60 g a day in men, alcohol is the dominant cause and the condition is treated as alcohol-related liver disease.

| Drink | Typical serving | Approximate alcohol |
|---|---|---|
| **Beer** | 330 ml bottle at about 5 per cent | about 13 g |
| **Beer (strong)** | 650 ml bottle at about 8 per cent | about 41 g |
| **Country liquor / arrack** | 60 ml | about 20-24 g, strength varies widely |
| **Whisky, rum, vodka, gin** | 30 ml (one standard peg) | about 10 g |
| **Whisky, rum, vodka, gin** | 60 ml (a large peg, or a restaurant pour) | about 20 g |
| **Wine** | 150 ml glass at 12 per cent | about 14 g |

Which produces an uncomfortable piece of arithmetic for anyone who drinks: two large pegs of whisky a day is roughly 40 g, which already puts a man in the MetALD band. And the 2024 European guidelines concluded that the old idea of a heart-protective small dose does not hold up in people with metabolic risk factors - the evidence for any protective effect in this group is inconsistent, while the evidence for harm to the liver is not. Chapter 20 deals with what to do about this honestly, including for people who are not going to stop.

CHAPTER 3

# Why this is an Indian problem: the numbers, the thin-fat body, your waist

Fatty liver is common everywhere rich food and sitting arrived together. But India's version has three features that make it different from the Western textbook description, and each one changes what you should do.

## The scale

The largest recent Indian dataset is the Phenome India cohort, published in The Lancet Regional Health - Southeast Asia in February 2026: 10,267 adults recruited through CSIR laboratories in 27 cities, of whom 7,764 had usable transient elastography. Of those, 3,712 met the criteria for MASLD - a crude prevalence of 47.8 per cent and an age-adjusted prevalence of 38.9 per cent. Nearly four in ten adults.

That figure sits in the middle of a range you will see quoted elsewhere, and it is worth knowing why the range is so wide. An AIIMS-led systematic review and meta-analysis put the pooled prevalence at 38.6 per cent in adults and 35.4 per cent in children. The MAP study, which looked at 13,750 people attending diabetes and endocrine clinics across 105 centres, found 68.2 per cent with MASLD on CAP measurement - but that population was already selected for metabolic disease. Government estimates used when India launched its fatty liver programme ranged from 9 to 32 per cent. Different populations, different machines, different definitions.

| Population | Share with MASLD | Source |
|---|---|---|
| **Indian adults, community-based, 27 cities** | 38.9 per cent (age-adjusted) | Phenome India cohort, Lancet Reg Health SE Asia, 2026 |
| **Indian adults, pooled analysis** | 38.6 per cent | AIIMS-led systematic review and meta-analysis, 2026 |
| **Adults attending diabetes and endocrine clinics** | 68.2 per cent | MAP study, 13,750 adults, 105 centres |
| **People already living with type 2 diabetes** | 60 to 85 per cent | India-specific consensus on MASLD in T2D, Diabetes Care 2025 |
| **Indian children, pooled** | 35.4 per cent (63.4 per cent in children with obesity) | AIIMS-led meta-analysis, 2026 |
| **Men versus women** | 45.9 per cent versus 33 per cent | Phenome India cohort, 2026 |

Those numbers multiply into the headline figure you may have read in the newspapers: roughly 120 million Indians estimated to be living with fatty liver, and about 20 per cent of liver transplants in India attributed to cirrhosis caused by it. A disease that is silent for two decades and then expensive for one is exactly the shape of public health problem that gets ignored until it is not.

## The thin-fat body

This is the part that catches people out. In the Phenome India data, 93.7 per cent of people with fatty liver were overweight or obese - but that means 6.3 per cent were not. Indian cohorts consistently show a phenotype that the West sees less often: normal BMI, normal-looking frame, and a liver full of fat, because the fat is inside the abdomen rather than under the skin. Muscle mass is lower, visceral fat is higher, and insulin resistance arrives at a body weight that would look fine on any Western chart.

INASL's own commentary notes that more than 85 per cent of Indian patients with "lean" NAFLD still meet MASLD criteria - because of central obesity or one of the other four metabolic risk factors. So if you are 68

kg and 5 feet 8 inches and have been told you cannot possibly have fatty liver, the person who told you that was using the wrong ruler.

### The three measurements that matter more than your weight

| Measurement | How to take it | Indian cut-off that matters |
|---|---|---|
| **Waist** | Standing, relaxed, tape at the level of the navel, at the end of a normal breath out. Do not suck in and do not pull the tape tight. | Above 90 cm in men, above 80 cm in women. |
| **Waist-to-height ratio** | Waist divided by height, both in the same unit. | 0.5 or above is the number most Indian guidance treats as central obesity - your waist should be less than half your height. |
| **BMI, read the Indian way** | Weight in kg divided by height in metres squared. | 23 or above is overweight, 25 or above is obese. The international cut-off of 30 is not the one to use for Indians. |

A 5 foot 6 inch (168 cm) man should therefore be aiming for a waist under 84 cm, not the 90 cm absolute cut-off, because half his height is 84. Most Indian men reading this will find their waist is 10 to 20 cm over. That single number, tracked monthly with a cheap tape measure, is a better progress marker than the weighing scale - because visceral fat responds to the changes in this book faster than body weight does, and you will see it move first.

CHAPTER 4

## The number that predicts your future: fibrosis, not fat

Every guideline in the world agrees on one thing about this disease: the amount of fat does not predict what happens to you. The amount of **scarring** does. This chapter is the reason the rest of the book is organised the way it is.

Scarring is staged F0 to F4. F0 is a normal liver. F1 is mild scarring, F2 is significant, F3 is advanced, F4 is cirrhosis. Fat without scarring is stage F0 to F1. The staging is what a hepatologist is actually trying to work out when they send you for a FibroScan.

| Stage | What it means | Typical time to advanced liver disease | Mortality risk versus F0-F1 |
|---|---|---|---|
| **F0-F1** | No or mild scarring | 22 to 26 years | 0.88 (no increase) |
| **F2** | Significant scarring | 9.3 years | 1.36 times |
| **F3** | Advanced scarring | 2.3 years | 2.54 times |
| **F4** | Cirrhosis | Already there | 5.19 times |

Read the middle column slowly, because it is the most useful set of numbers in this book. At F0-F1 you have two decades. At F2 you have about nine years - which sounds like a lot until you remember that nine years pass whether or not you do anything. At F3 you have roughly two years before the kind of trouble that puts people on a transplant list. This is why the difference between "fatty liver" and "fatty liver with F2 fibrosis" is not a detail; it is the whole story.

The reassuring half of the same data: the progression is not automatic. Most people with fat in the liver never reach cirrhosis. Estimates of how many progress vary with population and follow-up, but the consistent finding is that fibrosis stage, diabetes and continuing alcohol use are what separate the people who progress from the people who do not - and two of those three are things you can change.

## What actually kills people with fatty liver

Not liver failure, in most cases. The Indian health ministry's own programme documents make the point: cardiovascular disease is the most common cause of death in people with NAFLD. Fatty liver is a marker of a body-wide metabolic problem, and that problem damages arteries at least as efficiently as it damages the liver. This is why your doctor cares about your blood pressure and your LDL cholesterol when they are looking at your liver report, and why a book about the liver has to keep talking about the heart.

Practically, that means the plan in Part Three is not a liver diet. It is the diet that reduces visceral fat, blood pressure, triglycerides and blood sugar at the same time, which happens to be the diet that empties the liver. You are not choosing between your liver and your heart. There is only one plan.

> **So what is my actual number?**
>
> You can get a usable estimate of your fibrosis stage today, for free, from a blood report you probably already have. It is called FIB-4 and it needs your age, AST, ALT and platelet count. Chapter 6 walks through it with three worked examples. If you only do one thing after reading Part One, do that.

### Part One in five sentences

The word "fatty" on your ultrasound is a statement about brightness, not about danger, and the grade 1, 2 or 3 tells you nothing about scarring.

The condition is now called MASLD, and it is diagnosed by the presence of metabolic risk factors rather than by the absence of everything else.

Roughly four in ten Indian adults have it; most have never been given anything beyond a grade and a sentence about weight.

Fibrosis stage, not fat, predicts what happens to you: 22 to 26 years to trouble at F0-F1, about nine years at F2, about two at F3.

The most common cause of death in people with fatty liver is cardiovascular disease, which is why the plan in this book is the same plan your heart needs.

If you take one number out of Part One, take FIB-4. It is free, it takes four minutes, and it is the difference between watching and waiting.

PART TWO

# Reading your own numbers

Five chapters, one calculator, and the end of guessing what your report means.

CHAPTER 5

## Your LFT, line by line - and why a normal ALT proves nothing

The liver function test is the most-ordered blood test in India and the most-misread. It is not really a test of function at all - most of it measures enzymes that leak out of damaged liver cells, plus a few proteins and a platelet count. Here is what each line is telling you.

| Test | What it is | What a high or low value suggests in fatty liver |
|---|---|---|
| **ALT (SGPT)** | An enzyme inside liver cells. Leaks when cells are injured or inflamed. | Raised ALT is the classic signal of liver-cell injury. But normal ALT is common in people with fat in the liver, and INASL states plainly that transaminases cannot be used to tell simple fat from steatohepatitis. |
| **AST (SGOT)** | Similar enzyme, also found in muscle and red cells. | An AST that is higher than ALT, especially with a low platelet count, points towards more advanced scarring. Heavy alcohol use also pushes AST up more than ALT. |
| **GGT** | An enzyme of the bile ducts and liver, sensitive to alcohol and many medicines. | Often the first enzyme to rise in fatty liver, and often still raised when ALT has normalised. Very sensitive, not very specific. |
| **ALP** | Alkaline phosphatase, from bile ducts and bone. | A marked isolated rise suggests a bile-duct problem rather than fatty liver, and needs its own work-up. |
| **Bilirubin** | The yellow pigment the liver processes. | Raised bilirubin with yellow eyes or skin is a red flag - see Chapter 22. Mild isolated elevation is common and often harmless. |
| **Albumin and total protein** | Proteins the liver manufactures. | A falling albumin over time is a sign the liver's working capacity is reducing. This is one of the few genuine "function" measures on the panel. |
| **Platelet count** | Made in the bone marrow; falls when the spleen enlarges from portal pressure. | A platelet count drifting down is one of the earliest blood signs of advanced fibrosis. It is also an input to your FIB-4 score. |

### The four rules for reading your own LFT

1. **Compare with your own previous result, not with the lab's arrow.** The reference range is a statistical band, not a health target. If your ALT has gone from 24 to 44 over two years, that movement matters even though both numbers sit inside the printed range.
2. **A normal ALT does not clear you.** A large share of people with fatty liver - and a meaningful share of those with inflammation and scarring - have entirely normal enzymes. This is precisely why FIB-4 exists and why the international guidelines say not to use enzymes to decide who has the dangerous form.
3. **Look at the ratio.** ALT higher than AST is the usual pattern in fatty liver. AST higher than ALT, particularly with a platelet count under about 150, is the pattern that makes a doctor send you for elastography.

4 **One abnormal report is information; three abnormal reports over a year are a pattern.** Enzymes move with a fever, a hard workout, a wedding week of eating, a new medicine. Trends are the signal.

Two practical notes. First, get the actual numbers printed - a report that says "SGPT: 62 (H)" is enough to compute FIB-4, but a lab slip that only says "abnormal" is not. Second, if you are paying out of pocket, a liver function test is one of the cheapest useful tests in India: the CGHS notified rate for an LFT is Rs 425 at a non-NABH lab and Rs 500 at a NABH lab, and a whole-abdomen ultrasound is Rs 680 to Rs 800. Private labs vary, often by a factor of two, so ask before you book.

CHAPTER 6

## FIB-4 in four minutes: the free score that decides your next step

FIB-4 is a four-variable formula that estimates how much scarring you have. It was not designed for fatty liver specifically, but INASL's guidance calls it the most accurate of the free bedside scores, recommends it as the first fibrosis test at primary and secondary care level in India, and the 2025 India-specific consensus on MASLD in type 2 diabetes makes it part of routine screening. It costs nothing beyond a blood test you have probably already done.

**The formula**

**FIB-4 = (age in years x AST) / (platelet count x square root of ALT)**

Use AST and ALT in U/L exactly as printed on your report, and the platelet count in the units printed on Indian reports - cells per microlitre, usually written as 2.45 lakh or 245,000 or 245 x 10^9/L. If your report shows lakhs, write the number as 245, not 2.45. Then divide. Any phone calculator will do it; the square root button is the one with the radical sign.

### Three worked examples

| | Example A | Example B | Example C |
|---|---|---|---|
| **Age** | 42 | 55 | 63 |
| **AST** | 38 | 62 | 88 |
| **ALT** | 52 | 78 | 64 |
| **Platelets** | 245 | 168 | 96 |
| **The arithmetic** | (42 x 38) / (245 x 7.21) = 1596 / 1767 | (55 x 62) / (168 x 8.83) = 3410 / 1484 | (63 x 88) / (96 x 8) = 5544 / 768 |
| **FIB-4** | 0.90 | 2.30 | 7.22 |
| **What it means** | Low risk. No significant scarring likely. Manage the metabolic problem and repeat in a year. | Indeterminate. Scarring cannot be ruled out. Go for elastography. | High risk. Needs a hepatologist, not a repeat test in six months. |

### What your number means

| FIB-4 | Reading | What to do |
|---|---|---|
| **Below 1.0** | Significant fibrosis effectively ruled out in Indian cohorts - this is the cut-off INASL cites for Indian data. | You have time. Work on weight, waist, sugar and alcohol for 6 to 12 months, then repeat the blood test and recompute. |

| FIB-4 | Reading | What to do |
|---|---|---|
| **1.0 to 1.3** | Low risk, but not excluded. This is the band where most people with fatty liver in India actually sit. | Start the plan now. Repeat FIB-4 in 6 to 12 months. If anything else is abnormal - platelets falling, AST above ALT - ask for elastography sooner. |
| **1.3 to 2.67** | The grey zone. A real possibility of significant fibrosis that blood tests cannot settle. | Book a FibroScan. This is the single most useful thing you can do this month. |
| **Above 2.67** | High probability of advanced fibrosis. | See a hepatologist or gastroenterologist. Do not wait for the next annual check-up. |

Three cautions, because a number this easy to compute is also easy to over-trust. FIB-4 is less reliable at the extremes of age - under about 35 the formula tends to read low, and over 65 it tends to read high, so an older person can get a worrying score with a perfectly fine liver. It also leans heavily on the platelet count, which falls for reasons that have nothing to do with the liver, such as dengue, some medicines or a recent viral illness - so do not compute it during or just after an acute illness. And it estimates, it does not measure: a FIB-4 of 1.1 is not a guarantee, it is a probability that lets you plan.

Recompute it every time you get a blood count and an LFT. The trend in FIB-4 over two years tells you more than any single value: a score that stays at 1.2 is a stable liver, while a score that climbs from 1.1 to 1.9 is telling you something is progressing even though both numbers are individually unremarkable.

CHAPTER 7

## FibroScan: CAP, kPa, cost, and how to prepare so the reading is real

A FibroScan - technically vibration-controlled transient elastography, or VCTE - is the test that answers the question your ultrasound could not. It gives two numbers, and Indian labs print them on one line, which is why they confuse people.

| The number | Unit | What it measures | How to read it |
|---|---|---|---|
| **CAP** | dB/m (decibels per metre) | How much fat is in the liver - the same thing the ultrasound grade was gesturing at, but quantified. | Under 238 is normal. 238 to 260 is mild (S1), 261 to 292 moderate (S2), 293 and above severe (S3). These are the bands used in the Indian MAP study of 13,750 patients. |
| **LSM (liver stiffness)** | kPa (kilopascals) | How stiff - that is, how scarred - the liver is. This is the number that matters. | Under 6.0 rules out significant fibrosis. 8.2 and above rules it in. 13.6 and above suggests cirrhosis. INASL endorses the 8.2 and 13.6 thresholds for Indian practice. |

So a report that reads *"CAP 312 dB/m, LSM 6.4 kPa"* means: a lot of fat, no significant scarring. That is good news with a clear job attached - the fat is reversible, and the point is to reverse it before the stiffness number starts moving. A report reading *"CAP 271 dB/m, LSM 9.1 kPa"* means less fat but real scarring: that person needs a specialist opinion now, even though their ultrasound grade might look milder than the first person's.

### Getting a reading you can trust

Elastography is operator-dependent and meal-sensitive. Four things make a reading falsely high, and three of them are under your control on the day:

- **Eating before the scan.** A meal - especially a fatty one - temporarily raises liver stiffness. Fast for at least two to three hours, and longer if your appointment allows it.
- **Alcohol in the previous 24 hours.** Skip it. One dinner with drinks can move the number.

- **The probe used.** If your BMI is above 30, ask whether they are using an XL probe. Readings with the standard probe in a larger body are less reliable.
- **Other conditions.** Fluid in the abdomen, heart failure, a blocked bile duct or a very full stomach all distort the measurement. Tell the operator if any apply.

Ask for ten valid measurements and the median - a good report shows the number of measurements and the IQR (the spread). A wide spread means the reading is unreliable and should be repeated, whatever the median says. And do not repeat the scan every three months: fibrosis changes over years, not weeks. Twelve to twenty-four months is the sensible interval unless a specialist says otherwise.

### What it costs in India

| Where | Typical cost in 2026 | Notes |
|---|---|---|
| **Government hospital (AIIMS, Safdarjung and similar)** | Rs 200 to 500 | Cheapest by far. Wait times of days to weeks; CAP may not be available at every facility. |
| **CGHS empanelled centre** | Rs 978 (non-NABH) to Rs 1,150 (NABH) | The notified CGHS rate for FibroScan liver. Needs a CGHS card and a referral from a CGHS doctor. |
| **Standalone NABL diagnostic lab** | Rs 2,800 to Rs 5,000 with CAP | Usually same-day. Confirm CAP is included - it is sometimes charged as an add-on of Rs 1,000 to Rs 1,500. |
| **Private super-speciality hospital** | Rs 6,000 to Rs 9,000 | Often bundles a hepatologist review, which is worth something if the reading is borderline. |

One warning that saves money: FibroScan is an outpatient diagnostic, and most Indian health insurance policies do not pay for it unless you have an OPD add-on. Budget for it as cash. If the cost is the obstacle, the government-hospital route is not a compromise on quality - the machines are the same machines.

CHAPTER 8

## What else it could be: the tests that rule out the other causes

Under the new MASLD definition you no longer have to prove the absence of every other disease before the diagnosis can be made. But that does not mean other diseases are not there. INASL is explicit that at least hepatitis B and hepatitis C testing is prudent in every patient with elevated transaminases, and Indian consensus documents list a short set of additional tests when the picture does not fit.

| Test | What it rules out | When it is worth asking for |
|---|---|---|
| **HBsAg and anti-HCV** | Chronic hepatitis B and C | Once, for essentially everyone with fatty liver or raised enzymes. Both are common in India, both are treatable, and both change the whole plan. |
| **TSH** | Hypothyroidism | Once. An underactive thyroid worsens cholesterol and weight, and is easy to miss. |
| **HbA1c and fasting glucose** | Diabetes and prediabetes | Always. Two in three people with type 2 diabetes have fatty liver, and diabetes is what drives progression. |
| **Lipid profile** | High triglycerides, low HDL | Always - it is both a diagnostic criterion and a target. |
| **Ferritin and transferrin saturation** | Iron overload (haemochromatosis) | Ferritin is often mildly high in fatty liver itself; a high transferrin saturation is the more specific clue. |
| **Caeruloplasmin, and eye examination for a Kayser-Fleischer ring** | Wilson disease | Under 40 with unexplained liver disease. Rare, but catastrophic if missed, and treatable. |

| Test | What it rules out | When it is worth asking for |
|---|---|---|
| **ANA, anti-smooth-muscle antibody, IgG, AMA** | Autoimmune hepatitis and primary biliary cholangitis | Women with persistently raised enzymes, or anyone whose picture does not fit. |
| **Anti-tTG (coeliac)** | Coeliac disease | Unexplained enzymes with anaemia, bloating or weight loss. Indian consensus work-up includes it. |
| **Alpha-1 antitrypsin** | Alpha-1 antitrypsin deficiency | Young patients, or family history of early emphysema or liver disease. |
| **A frank alcohol history, or a biomarker such as GGT or CDT** | Alcohol-related liver disease | Where drinking is more than the MASLD threshold - see Chapter 2 for the gram cut-offs. |

Two non-disease causes are worth naming because they are common and completely reversible. The first is **medicines**: steroids, methotrexate, tamoxifen, amiodarone, valproate and the anti-tuberculosis combination all affect the liver, and the last of those is the single largest cause of drug-induced liver injury in India. The second is **supplements and herbal products**, which Chapter 18 deals with at length - including the uncomfortable finding that traditional and alternative medicines accounted for 14 per cent of 1,288 cases in the Indian Network of DILI, and that cases linked to them tend to be severe.

If you have been given a diagnosis of fatty liver and none of this testing was ever done, that is not a reason to panic. It is a reason to take a printed list to your next appointment and ask which of these are worth doing in your case. A reasonable minimum for most people with fatty liver in India is: HbA1c, fasting glucose, lipid profile, TSH, HBsAg, anti-HCV, and a platelet count with AST and ALT - which together also give you your FIB-4.

CHAPTER 9

## Build your own risk map

You now have four separate pieces of information, and they do not always agree. One person has grade 3 fat and no scarring; another has grade 1 fat and F2 fibrosis. This chapter puts them on one page so you can see your own situation whole - and so you can tell, twelve months from now, whether you are winning.

Fill in Worksheet W1 at the back of the book as you read this. There are four rows, and each one answers a different question.

| Row | The question it answers | What goes in it |
|---|---|---|
| **1. Fat** | How much fat is in the liver? | Ultrasound grade (1/2/3) and, if you have it, CAP in dB/m. |
| **2. Inflammation** | Is the liver being injured right now? | ALT, AST, GGT as printed. Note whether AST is higher than ALT. |
| **3. Fibrosis** | Is there scarring, and how much? | FIB-4 number, and LSM in kPa if you have had elastography. |
| **4. Metabolic load** | What is driving all of it? | Weight, waist, BMI read with Indian cut-offs, HbA1c, blood pressure, triglycerides, HDL, and alcohol in grams per day. |

### Then find your lane

| Lane | You are here if | What to do this month |
|---|---|---|
| **Green** | FIB-4 under 1.0, normal or mildly raised enzymes, no diabetes, waist under the cut-off. | The twelve-week plan in Part Three, a repeat blood test in 6 to 12 months, and a repeat ultrasound in about a year. Nothing urgent, but nothing to postpone either. |

| Lane | You are here if | What to do this month |
|---|---|---|
| **Amber** | FIB-4 between 1.0 and 1.3 with diabetes or a waist well over the cut-off; or FIB-4 1.3 to 2.67; or enzymes that have been raised for more than six months. | Book elastography. Start the plan now. Get HbA1c and lipids controlled. See a physician within a month with the printed reports. |
| **Red** | LSM 8.2 kPa or above; or FIB-4 above 2.67; or platelets below 150; or albumin falling; or bilirubin raised with jaundice. | Hepatologist or gastroenterologist, not a general physician, and not next quarter. Take every report you have. Chapter 22 lists the symptoms that mean go today. |

### A note for the person who feels fine

Almost everyone in this book feels fine. Fatty liver has no symptoms until it has been quietly working for twenty years, which is exactly why it gets ignored and exactly why a two-hour investment in understanding your own numbers is worth making now. The people who end up on a transplant list are not the ones who felt worst at the start. They are the ones who were never told what to look at.

### Part Two in five sentences

Your LFT is a set of leaked enzymes and a platelet count, not a measure of how well the liver works - and a normal ALT does not exclude inflammation or scarring.

FIB-4, computed from age, AST, ALT and platelets, is the free first-line test that Indian guidance recommends, and the trend over two years matters more than any single value.

FibroScan gives two numbers: CAP for fat and LSM in kPa for scarring. The kPa is the one that decides your next step.

MASLD is no longer a diagnosis of exclusion, but hepatitis B, hepatitis C, thyroid disease and a frank alcohol history still have to be checked.

Four rows - fat, inflammation, fibrosis, metabolic load - put your whole situation on one page, and the lane you land in tells you whether this is a twelve-week project or a specialist referral.

Fill in Worksheet W1 before you read Part Three. A plan built on numbers you actually have is a different thing from a plan built on worry.

PART THREE

# The twelve-week reversal plan

Weight, plate, movement, sleep. The four levers that actually move liver fat.

CHAPTER 10

## Your weight target: the 3 per cent, 7 per cent and 10 per cent rules

Weight loss is the only intervention with decades of evidence behind it for fatty liver, and the guidelines are unusually precise about how much. The numbers below are consistent across the European 2024 guidelines, AASLD guidance and the Indian consensus documents.

| Weight lost | What it changes | Timeframe the guidance suggests |
|---|---|---|
| **3 to 5 per cent of body weight** | Reduces liver fat measurably. In people with a normal BMI, 3 to 5 per cent can produce remission of MASLD. | Achievable in 8 to 12 weeks |
| **7 per cent or more** | Improves inflammation and the activity of steatohepatitis - the part that injures cells. | 3 to 6 months |
| **10 per cent or more** | The level at which fibrosis itself has been shown to improve. | 6 to 12 months |
| **7 to 10 per cent over 6 months** | The target the Indian consensus recommends as the standard goal. | 6 months |

Now do the arithmetic for yourself, because a percentage is not a plan. If you weigh 82 kg, 5 per cent is 4.1 kg and 7 per cent is 5.7 kg. If you weigh 68 kg, 5 per cent is 3.4 kg. Write your number on Worksheet W3. Then divide by twelve: 5 per cent in twelve weeks means roughly 350 to 500 grams a week for most Indian adults. That is a completely unremarkable rate of loss - half a kilo a week, which is what a modest calorie deficit produces - and it is enough to change a scan.

### What not to do

- **Do not crash diet.** Very rapid weight loss, particularly with prolonged fasting or a liquid-only regimen, can transiently worsen inflammation and gallstone risk. Slow and boring is the correct speed.
- **Do not aim for the scale alone.** Track waist monthly. Visceral fat responds first, and the tape measure will show progress in weeks when the scale is stuck.
- **Do not try to lose weight you do not have.** If your BMI is 22 and your waist is 88 cm, your target is not 8 kg of weight loss - it is 4 cm off the waist, more protein, more muscle and better sleep. The European guidance is explicit that a 3 to 5 per cent loss is the right target in normal-weight MASLD.
- **Do not expect linear progress.** Weight loss plateaus. A fortnight of no change after six good weeks is normal physiology, not failure. Change one variable and continue.

Two things worth knowing about the ceiling. Bariatric surgery is a legitimate, guideline-endorsed option for people with MASLD and severe obesity - it improves liver histology - and it is not a confession of weakness. And for people with diabetes or obesity, the GLP-1 class of medicines now has a liver indication in India; that is Chapter 17, and it is worth reading before you assume medicine is not for you.

CHAPTER 11

# The Indian thali, rebuilt: what ICMR-NIN 2024 actually recommends

Indian dietary advice for fatty liver usually arrives as a photocopy: no rice, no roti after 7 pm, no mango, drink karela juice. The actual national guidance is more specific and much less punitive. The ICMR-National Institute of Nutrition's Dietary Guidelines for Indians 2024 sets out a plate, and it is a plate you can eat for the rest of your life.

## The numbers in the 2024 guidelines

| What | The recommendation | What it means at the table |
|---|---|---|
| **Cereals and millets** | No more than 45 per cent of daily energy; about 250 g raw per day for a 2,000 kcal diet | Roughly 4 to 5 rotis or 2 to 3 small bowls of rice a day, not at every meal and not in every snack. |
| **Whole grains and millets** | At least 50 per cent of cereals as whole grain; 20 per cent of cereals from millets | Atta over maida. One meal a day with a millet - jowar, bajra, ragi, little millet - is enough to hit the millet target. |
| **Pulses, beans, eggs, flesh foods** | About 14 to 15 per cent of energy; 85 g raw pulses per day | A proper katori of dal at lunch and dinner, not a ladle of gravy. This is the change most Indian plates need most. |
| **Vegetables and greens** | 400 g a day, including green leafy vegetables | Half the plate. Cooked sabzi plus a raw salad, both. |
| **Fruit** | 100 g a day, fresh, not juiced | One small fruit. Fruit is not the enemy; fruit juice is. |
| **Milk and curd** | 300 ml a day | A cup of curd with lunch and a glass of milk, or equivalent. |
| **Nuts and oilseeds** | About 35 g a day | A small handful - 4 to 6 almonds, 2 walnuts, a spoon of seeds. |
| **Cooking oil and ghee** | 25 to 30 g a day in total | About 5 to 6 teaspoons for the whole day, including what goes into tadka and what is poured at the table. |
| **Added sugar** | Under 5 per cent of energy, about 20 to 25 g a day; none for children under 2 | Two teaspoons of sugar in tea is 8 to 10 g. Two cups of sweet tea plus a biscuit uses up the day's allowance. |
| **Salt** | 5 g a day maximum | One level teaspoon for the whole day, including pickles, papad, namkeen and packaged food. |

The guideline also makes a point that matters more than any single number: as a country we eat too much cereal and too little of everything else. Cereals supply 50 to 70 per cent of Indian energy intake against a recommendation of 45, while pulses, meat, poultry and fish together supply 6 to 9 per cent against a recommendation of about 14. The fatty liver diet, in Indian terms, is mostly the job of moving food from the cereal column into the pulse and vegetable columns.

## Three rules that beat any diet plan

1. **Total calories decide whether the liver empties; food quality decides how easy it is to keep going.** The Indian consensus is blunt about this: a 30 per cent reduction in calories, or 500 to 1,000 kcal a day less, matters more than which named diet you follow. The Mediterranean pattern is recommended because it is proven, but a local diet that cuts both refined carbohydrate and excess oil, built from foods you actually like, is the one that survives month four.

2. **Protein at every meal.** Dal, curd, paneer, eggs, fish, chicken, sprouts, chana. Protein keeps you full, protects muscle while you lose fat, and displaces cereal. The 2024 guidelines put 10 to 15 per cent of energy from protein; most people with fatty liver do better at the upper end of that.
3. **Eat in a window, not all day.** The guidelines recommend restricting meals to two or three times a day and not snacking in between. Three meals and no constant chai-and-biscuit grazing is a bigger change than most people expect, and it is the one that reduces total intake without counting anything.

A word on ghee, because it comes up in every conversation. The guideline's issue is total visible fat, not ghee specifically: 25 to 30 g of oil and ghee combined per day. One teaspoon of ghee on two rotis is about 10 g and fits. Half a cup poured over dal does not. Nothing about fatty liver requires you to give up ghee, rice, eggs, fruit or coffee - it requires you to stop exceeding the day's total, and to move the balance of the plate.

CHAPTER 12

## The five things to cut first, and the ones not worth bothering with

If you change ten things at once you will change nothing by month two. Cut in this order. Each of these five has direct evidence behind it for liver fat, and each one is a single decision rather than a daily act of willpower.

1. **Sugar-sweetened drinks and packaged fruit juice - completely.** This is the highest-yield change in the book. In a controlled study, six months of daily sugar-sweetened beverage intake raised liver fat by 132 to 143 per cent, along with visceral fat and triglycerides. The fructose in these drinks goes straight into new fat production in the liver. Replace with water, unsweetened buttermilk, nimbu pani without sugar, or plain soda with lime. Do not replace with diet cola and assume the problem is solved - the evidence on artificial sweeteners is not reassuring either.
2. **Deep-fried snacks.** Samosa, kachori, pakora, puri, namkeen, chips, the entire evening-snack economy. Not because frying is evil but because a single serving can carry 10 to 15 g of oil, which is half your day's allowance before dinner is served. Roasted chana, makhana, a boiled egg, a handful of peanuts - all replacements that do not feel like punishment.
3. **Refined cereal volume.** Not rice and roti as foods, but the quantity and the refinement: maida bread, white rice at all three meals, instant noodles, biscuits, rusk, white pasta. Swap to atta, keep rice to one meal, and put a millet in once a day.
4. **Mithai and bakery as a daily habit.** Festivals are festivals; Wednesdays are not. The 2024 guideline's 20 to 25 g added sugar limit is roughly one small sweet. If you have a sweet tooth, move it to a fixed day of the week rather than trying to eliminate it.
5. **Alcohol, and late-night eating.** These are grouped because they are the same behaviour in different clothes: the last three hours of the day, when the day's discipline is exhausted. Both independently worsen liver fat. Chapter 20 handles alcohol properly; for late eating, the rule is a three-hour gap between your last food and bed.

### And the things not worth bothering with

| The advice you will hear | The honest answer |
|---|---|
| **"Give up rice completely."** | Rice is not the problem; the quantity, the number of times a day, and what it is eaten with are. A small bowl of rice with dal and sabzi at lunch fits easily inside 45 per cent of energy from cereals. |
| **"Give up eggs, they have cholesterol."** | Two eggs a day do not raise LDL meaningfully in most people, and they are one of the cheapest sources of the protein your plate is short of. |

| The advice you will hear | The honest answer |
|---|---|
| **"Drink karela / methi / jamun juice to clean the liver."** | No controlled trial shows these change liver fat. Bitter gourd has a modest effect on blood sugar in some studies; that is a different claim from reversing steatosis. Drink them if you like them, not as treatment. |
| **"Coffee is bad for the liver."** | The opposite. Three or more cups of coffee a day is consistently associated with less advanced liver disease in the guidelines, and coffee appears in the EASL, AASLD and APASL recommendations. The sugar and the cream are the problem, not the coffee. |
| **"Fruit has sugar, avoid it."** | Whole fruit at 100 g a day is explicitly recommended. The fructose in fruit arrives with fibre and water; the fructose in a soft drink does not. |
| **"Do a liver detox / cleanse."** | The liver detoxifies itself. There is no product that has been shown to remove liver fat, and some 'liver' products have caused the very injury they claim to prevent - Chapter 18. |
| **"Fast for 16 hours and it will reverse."** | Intermittent fasting may help some people control total intake, and the guidance says it can be considered and individualised. It is not yet established as a treatment in its own right, and it does not beat a sustained calorie deficit. |

CHAPTER 13

# Six breakfasts, ten lunches and dinners, one seven-day menu

This chapter is deliberately concrete, because "eat a balanced diet" has never changed anyone's scan. Portions below assume an adult aiming at roughly 1,600 to 1,800 kcal a day with 5 to 6 teaspoons of oil total. If you are taller, more active, or male, add a roti or an extra katori of dal rather than adding oil.

## Six breakfasts, rotated

| Breakfast | The plate | Why it works |
|---|---|---|
| **1. Poha with peanuts and sprouts** | 1.5 cups poha cooked with 1 tsp oil, a handful of roasted peanuts, 1/2 cup moong sprouts, curry leaves, lemon. One boiled egg or 100 g curd on the side. | Flattened rice cooks fast, the peanuts and egg add protein, and the total oil stays low. |
| **2. Besan chilla, two** | 2 chillas made from 60 g besan with onion, tomato, coriander, 1 tsp oil total. Mint chutney without sugar. 150 ml milk or curd. | High protein, low glycaemic load, and it holds you until lunch. |
| **3. Oats with curd and seeds** | 40 g rolled oats soaked overnight in 150 g curd, 1 tbsp chia or flax, a few walnuts, half a chopped guava. No sugar; cinnamon if you need sweetness. | Soluble fibre slows glucose absorption; the seeds and curd carry the protein. |
| **4. Idli with sambar** | 3 idlis, 1 bowl sambar with extra dal and vegetables, coconut chutney measured to 1 tbsp. | Fermented, low in oil, and the sambar is a protein delivery system if you make it thick. |
| **5. Two eggs, one roti, sabzi** | 2 eggs - boiled, or bhurji with 1 tsp oil - 1 atta roti, and the previous night's leftover dry sabzi. | The most satiating breakfast on the list, and the fastest to make. |
| **6. Daliya with vegetables** | 50 g broken wheat cooked with carrot, peas, beans and 1 tsp ghee, with 100 g curd. | Whole grain, high fibre, and it does not spike glucose the way upma made with rava does. |

What to stop doing at breakfast: two biscuits with tea and nothing else, which is not a meal and sets up a mid-morning samosa; and a glass of packaged juice with cornflakes, which is a dessert with a health claim.

## Ten lunches and dinners

Each of these is a full plate: about 1 katori of dal or a protein, 1 to 1.5 katoris of cooked vegetable, 2 rotis or 1 small bowl of rice, and a salad or curd. Cook with 1 to 1.5 teaspoons of oil per person per dish.

| | The dish | The protein |
|---|---|---|
| **1** | Rajma with brown rice and a cucumber-tomato salad | Rajma |
| **2** | Palak paneer (80 g paneer, low oil) with 2 rotis | Paneer |
| **3** | Grilled or tandoori chicken with a big kachumber and one roti | Chicken |
| **4** | Chana masala with 1 bowl rice and bhindi sabzi | Chana |
| **5** | Moong dal khichdi with curd and a side of lauki sabzi | Dal plus curd |
| **6** | Fish curry (rohu, surmai, or any local fish) with rice and beans poriyal | Fish |
| **7** | Mixed dal tadka, baingan bharta (1 tsp oil), 2 jowar rotis | Dal |
| **8** | Soya chunk and vegetable curry with 2 rotis and salad | Soya |
| **9** | Sambar with drumstick and pumpkin, 1 bowl rice, cabbage poriyal, curd | Dal |
| **10** | Egg curry, 2 eggs, with 1 roti and a large salad | Eggs |

## One week, laid out

| Day | Breakfast | Lunch | Dinner | Snack |
|---|---|---|---|---|
| **Mon** | Poha, peanuts, egg | Rajma, brown rice, salad | Moong khichdi, curd, lauki | Roasted chana |
| **Tue** | Besan chilla x2 | Chicken, kachumber, 1 roti | Palak paneer, 2 rotis | Buttermilk, 5 almonds |
| **Wed** | Oats, curd, seeds | Chana masala, rice, bhindi | Fish curry, small rice, beans | Guava |
| **Thu** | Idli x3, sambar | Dal tadka, 2 jowar rotis, bharta | Soya curry, 2 rotis, salad | Makhana |
| **Fri** | Eggs, 1 roti, sabzi | Sambar rice, poriyal, curd | Grilled chicken, big salad | Peanuts |
| **Sat** | Daliya with vegetables | Egg curry, 1 roti, salad | Rajma, 1 roti, cucumber raita | Fruit |
| **Sun** | One favourite breakfast, portioned | Eat out - see the rules below | Light: dal, 2 rotis, sabzi | Nothing after 8 pm |

## Eating out without wrecking the week

- **Biryani:** one serving, not two; ask for raita instead of extra salan; leave the oil at the bottom of the plate - it is genuinely most of the difference.
- **South Indian:** idli, dosa and uttapam are fine. The danger is the chutney and sambar refilled four times, and the oil the dosa is cooked in. Ask for less oil, and skip the potato masala.
- **North Indian restaurant:** tandoori and tikka over creamy curries; roti over naan or paratha; dal tadka over dal makhani. One gravy dish shared between two is a portion, not a deprivation.
- **Chaat and street food:** choose the ones that are not fried - bhel with less sev, fruit chaat, grilled sandwiches without cheese and butter. Avoid pani puri as a meal; it is fine as a treat.
- **Festival days:** eat one sweet, at the meal, not five across the day, and cut the cereal portion at that meal rather than cutting the sweet. This is how the guideline's sugar limit is meant to work.

CHAPTER 14

# Movement: 200 minutes a week, and the walk after dinner

Exercise improves liver fat partly through weight loss and partly independent of it - which means that even in a month where the scale does not move, the liver is still benefiting. The Indian consensus recommendation is specific: 30 to 45 minutes of aerobic activity on at least five days a week, roughly 200 minutes, at 60 to 70 per cent of your maximum heart rate. The international baseline is 150 to 300 minutes of moderate activity a week. Aim for the higher end if you can.

A simple way to find 60 to 70 per cent of maximum heart rate without a watch: you should be able to talk in short sentences but not sing. Brisk walking that makes you slightly warm and slightly out of breath. If you can chat comfortably the whole way, walk faster.

| Week | Aerobic | Strength | The habit you are building |
|---|---|---|---|
| **1-2** | 15 minutes brisk walk, 5 days | None yet | Showing up at the same time daily |
| **3-4** | 20 minutes, 5 days; add a 10-minute walk after dinner | 2 days: squats, wall push-ups, chair-supported sit-to-stand, 2 sets each | The after-dinner walk becomes automatic |
| **5-6** | 25 to 30 minutes, 5 days; one longer 40-minute walk at the weekend | 2 days, 3 sets; add a resistance band row and a light dumbbell press | Walking gets faster without effort |
| **7-8** | 30 minutes, 5 days; add 2 minutes of faster walking every 10 minutes | 2 to 3 days | You start missing it on rest days |
| **9-10** | 35 minutes, 5 days, or 25 minutes of cycling or swimming | 3 days | Exercise stops being a decision |
| **11-12** | 40 minutes, 5 days - about 200 minutes a week | 3 days | This is now your baseline, not your programme |

Three notes that make the difference between a plan and a good intention. **Resistance training is not optional** - it protects muscle while you lose fat, and muscle is where glucose goes. Two sessions of twenty minutes with bodyweight and a resistance band is enough to start. **The post-meal walk is the cheapest win available**: ten to fifteen minutes after your largest meal blunts the glucose spike, and it is easier to sustain than a gym membership. And **sitting is its own risk factor** - if your work is a desk and a phone, stand up for two minutes every forty minutes; the total step count matters less than not being still for eight hours.

> **If you have knee pain, or you are 65 and have not exercised in years**
>
> Start with water walking or a stationary cycle, both of which spare the knees, and add chair-based strength work. Twenty minutes is enough to begin. If you have known heart disease, uncontrolled blood pressure, or you get chest pain or unusual breathlessness on exertion, get cleared by a doctor before you increase intensity - not because exercise is dangerous, but because the first month of a new programme is the wrong time to discover a cardiac problem.

CHAPTER 15

# Sleep, stress, shifts, and the apnoea you may not know you have

There is a reason the plan has a sleep chapter. Short sleep and untreated sleep apnoea both worsen insulin resistance and drive appetite, and obstructive sleep apnoea is strongly associated with fatty liver - the two conditions share the same visceral fat. A person sleeping five hours a night and snoring loudly is fighting the diet with one hand tied behind their back.

## The two-minute apnoea screen

The STOP-BANG questions are the standard quick screen. Count one point for each yes:

1. **S** - Do you snore loudly, loud enough to be heard through a closed door?
2. **T** - Do you often feel tired or sleepy during the day?
3. **O** - Has anyone observed you stop breathing, or gasping or choking, during sleep?
4. **P** - Do you have high blood pressure, or take medicine for it?
5. **B** - Is your BMI above 35?
6. **A** - Are you over 50 years old?
7. **N** - Is your neck circumference more than 40 cm?
8. **G** - Are you male?

Three or more points is worth a conversation with a doctor about a sleep study. Five or more is worth having it soon. Treating apnoea - with weight loss, positional measures, or CPAP where indicated - improves blood pressure, daytime function and metabolic control, and it is one of the most under-diagnosed conditions in Indian adults with a large waist.

## The rest of the night

- **Seven hours, at a consistent time.** Consistency matters as much as duration for glucose control.
- **Shift work is a genuine metabolic load.** If you work nights, protect a fixed sleep block, keep the main meal before the shift rather than at 3 am, and avoid the sugary snacks that keep night shifts running. Ask for a yearly HbA1c and lipid profile - shift workers are under-screened.
- **Stress eating is not a character flaw.** Cortisol raises appetite for exactly the foods on the cut list. Ten minutes of walking, or breathing exercises, before the meal you tend to overeat is more effective than trying to want less.
- **Yoga: helpful, but not yet proven for fibrosis.** Indian guidance is honest about this - yoga improves flexibility, stress and sometimes weight, and the effect specifically on MASLD needs more research. Do it for the reasons it works, and count it towards your 200 minutes if it is vigorous enough to make you warm.

CHAPTER 16

# The twelve-week tracker

Twelve weeks is long enough for the liver to change and short enough to hold in your head. The table below is the whole plan compressed into one page - copy it into your phone, or use Worksheet W3 at the back. Tick a box only if you actually did it; an honest row of three ticks beats a dishonest row of seven.

| Week | The one focus for this week | Weight | Waist | Walk x5 | No sugary drinks | 3 meals, no grazing | Sleep 7h |
|---|---|---|---|---|---|---|---|
| 1 | Measure everything: weight, waist, photos of your reports | | | | | | |
| 2 | Cut sugar-sweetened drinks to zero | | | | | | |
| 3 | Add protein to breakfast every day | | | | | | |
| 4 | 15-minute walk after your largest meal | | | | | | |
| 5 | Half the plate vegetables, twice a day | | | | | | |
| 6 | Two strength sessions this week | | | | | | |
| 7 | Review: has the waist moved? Repeat blood test if due | | | | | | |
| 8 | One millet meal a day | | | | | | |
| 9 | No food within three hours of bed | | | | | | |
| 10 | Eat out once, using the rules in Chapter 13 | | | | | | |
| 11 | Increase walking pace - you should be slightly breathless | | | | | | |
| 12 | Full review: weight, waist, energy, sleep. Book the follow-up | | | | | | |

At the end of twelve weeks you are not finished - you are calibrated. The point of the tracker is not the ticks. It is that on day 84 you will know, from your own measurements, which of these four levers your body responds to. Most people find it is two of them, and it is rarely the two they expected.

### Part Three in five sentences

Losing 3 to 5 per cent of body weight reduces liver fat; 7 per cent improves inflammation; 10 per cent is where fibrosis itself starts to improve. Half a kilo a week is the right speed.

The ICMR-NIN 2024 plate is not a punishment: cereals down to 45 per cent of energy, pulses and vegetables up, oil at 25 to 30 g, added sugar under about 25 g, salt 5 g.

Cut sugar-sweetened drinks, deep-fried snacks, refined cereal volume, daily sweets and late-night eating - in that order, one at a time.

Two hundred minutes a week of brisk activity, two strength sessions and a fifteen-minute walk after your largest meal will change your liver whether or not the scale moves.

Sleep and untreated apnoea are part of the disease, not a lifestyle footnote.

The plan fails on week five, not week one. Decide now what you will do on the day you do not feel like it.

PART FOUR

# Medicines, supplements and traps

What is approved in India, what is being sold to you, and what to do about both.

CHAPTER 17

## What is actually approved in India in 2026

This is the chapter most people skip to, and it needs one sentence of framing before the table: **none of these medicines is for simple fatty liver.** Every approval below is for steatohepatitis - fat plus inflammation - usually with significant fibrosis. If your FIB-4 is 0.9 and your LSM is 5 kPa, the correct treatment is Part Three of this book, and a drug would be an expensive way of avoiding the actual work.

The second sentence of framing: in every one of the trials below, the placebo group also improved substantially, because everyone received diet and exercise counselling. In the semaglutide trial, a third of the placebo group resolved their steatohepatitis. Drugs add to lifestyle change. They do not replace it, and no regulator has approved one that does.

| Medicine | Status in India (November 2026) | What the trials showed | Who it is for |
|---|---|---|---|
| **Saroglitazar 4 mg once daily (PPAR alpha/gamma agonist)** | Approved by the Drug Controller General of India for NASH with F1 to F3 fibrosis in March 2020, and for NAFLD with obesity, type 2 diabetes, dyslipidaemia or metabolic syndrome in December 2020. Available only in India. | In the EVIDENCES II trial of 102 patients with biopsy-proven NASH, 52.3 per cent on saroglitazar achieved a two-point or greater fall in the activity score without worsening of fibrosis, against 23.5 per cent on placebo, at 52 weeks. A real-world Indian study of 91 patients found enzymes improved, but CAP and stiffness improved only in those who also lost at least 5 per cent of body weight. | Non-cirrhotic NASH or NAFLD with metabolic comorbidity, prescribed by a doctor. Minimum course in the guidance is one year. Not for cirrhosis. |
| **Semaglutide 2.4 mg weekly injection (GLP-1 receptor agonist)** | Approved by CDSCO in July 2026 for non-cirrhotic MASH with moderate to advanced fibrosis (F2-F3), alongside a reduced-calorie diet and increased physical activity. The first and only GLP-1 receptor agonist approved for MASH in India. | In the ESSENCE trial (1,197 patients, interim analysis of 800 at 72 weeks): steatohepatitis resolved in 62.9 per cent versus 34.3 per cent on placebo; fibrosis improved by at least one stage in 36.8 per cent versus 22.4 per cent; average weight loss 10.5 per cent versus 2.0 per cent. | Adults with confirmed F2-F3 fibrosis, usually with obesity or type 2 diabetes. Not approved for cirrhosis, or for simple steatosis without fibrosis. Gastrointestinal side effects are common at the start. |
| **Resmetirom (oral thyroid hormone receptor beta agonist)** | Approved by the US FDA in March 2024. **Not marketed in India** as of November 2026; an Indian phase 3 trial is running and a launch is projected for some years after patent expiry. | In MAESTRO-NASH (966 patients, 52 weeks): MASH resolved in 25.9 to 29.9 per cent versus 9.7 per cent on placebo; fibrosis improved in 24.2 to 25.9 per cent versus 14.2 per cent. | Not currently an Indian option. If you see it advertised online, that is a red flag. |
| **Pioglitazone 30 mg daily** | Available and cheap in India; approved for diabetes. Used off-label for MASH, and recommended for it by INASL in both diabetic and non-diabetic adults. | In PIVENS, pioglitazone significantly increased resolution of steatohepatitis compared with placebo, with reduction in steatosis and inflammation but no clear effect on fibrosis. | People with MASH, particularly with diabetes or prediabetes. Causes weight gain, and raises osteoporosis risk in postmenopausal women - a real trade-off to discuss. |

| Medicine | Status in India (November 2026) | What the trials showed | Who it is for |
|---|---|---|---|
| **Vitamin E 800 IU daily** | Widely available over the counter. INASL recommends it for biopsy-proven NASH in non-diabetic adults; the 2024 European guidelines and the 2025 global consensus do *not* recommend it as a MASH therapy because the evidence is not robust enough to outweigh the risks. | In PIVENS, vitamin E improved steatosis, inflammation and ballooning, but not fibrosis. | A shrinking role, and one to decide with a doctor. Long-term use has been associated with an increased risk of prostate cancer and haemorrhagic stroke. |

### What is not recommended, and is often sold anyway

Indian consensus documents list the drugs that have been used without substantial supportive evidence for continued use in MASLD: ursodeoxycholic acid, omega-3 fatty acid supplements, silymarin (milk thistle) and orlistat. Obeticholic acid was studied and is not recommended. If a doctor prescribes one of these, ask what it is expected to change and how you will know whether it worked. If a shop sells it to you as a liver cure, walk away.

### If you have diabetes or obesity, the order changes

The most useful medicine decision for many people with fatty liver is not a liver drug at all - it is choosing the right diabetes or weight medicine. GLP-1 receptor agonists and pioglitazone have liver-relevant benefits; SGLT2 inhibitors and metformin are reasonable for glucose control but are not MASH-targeted therapies. If you already take insulin or a sulfonylurea and your weight is climbing, that is a conversation worth having with your physician, not a reason to stop anything on your own.

CHAPTER 18

## The supplement trap: liver tonics, detox kits and the DILI numbers

This chapter will annoy somebody, so let us start with the evidence rather than the opinion. India has a national registry of drug-induced liver injury - the Indian Network for DILI, or INDILI. In its series of 1,288 patients, the single largest cause was the anti-tuberculosis combination at 46 per cent. The second largest was traditional and alternative medicines, at 14 per cent. Mortality from drug-induced liver injury in India is reported at 12 per cent in the national registry and 17 per cent in a large single-centre series; when drug-induced injury causes acute-on-chronic liver failure, mortality reaches 47 per cent.

Now the part that matters for a person with fatty liver specifically. A single-centre series from southern India looked at 94 patients with severe liver injury who had taken Ayurvedic and herbal medicines; 33 met causality criteria for herbal drug-induced injury, and five of them died - 18.5 per cent. Analysis of the retrieved products found arsenic and mercury in a significant number, and the presence of those heavy metals was significantly associated with death. Another series found that among patients who already had cirrhosis and were taking these products, 35.7 per cent presented with severe drug-induced injury, with 53 per cent mortality.

### The specific problem with "liver" products

The products most likely to be sold to you are exactly the ones with the least evidence and, in some cases, the most documented harm. The published case literature implicates, among others: ashwagandha (prolonged cholestatic hepatitis, sometimes taking months to resolve), guggul, giloy (Tinospora cordifolia), high-dose turmeric supplements, bakuchi, senna and aloe. Note what is *not* being said here: these are not poisons, and a teaspoon of haldi in your dal is not the issue. The risk concentrates in concentrated extracts, in unlabelled polyherbal mixtures with twenty or fifty ingredients, in products bought online, and in anything whose label does not list what is in it.

| What is being sold | The claim | The evidence |
|---|---|---|
| **"Liver tonic" syrups and Liv-type proprietary products** | Protects and cleanses the liver | No product in this category has been shown to reduce liver fat or fibrosis in a controlled trial. Proprietary Ayurvedic formulations are the second most common culprit category in Indian herbal liver-injury series. |
| **Milk thistle / silymarin capsules** | Repairs liver cells | Listed by Indian consensus as lacking substantial supportive evidence for continued use in MASLD. Not harmful for most people; also not doing anything measurable. |
| **TUDCA, NAC, alpha-lipoic acid** | Detox, antioxidant | No controlled evidence for steatosis or fibrosis. NAC has a role in paracetamol overdose, which is a different situation entirely. |
| **"Fat burner" and gym supplements** | Burns abdominal fat | The most consistently dangerous category after unlabelled herbals - multi-ingredient proprietary blends with no declared quantities, and a documented record of severe liver injury. |
| **Homeopathic liver drops** | Corrects liver function | No mechanism, no evidence, and dilution to the point that any effect is placebo. Harmless in itself; harmful when it replaces something that works. |
| **"Liver detox" teas, juices, enemas** | Flushes toxins | The liver does not need flushing. Some herbal teas contain senna or anthraquinones that cause diarrhoea and electrolyte loss, which is actively risky if you have cirrhosis. |

> **The three-month supplement holiday**
>
> If you are currently taking anything that is not prescribed - a liver tonic, a herbal mixture, a fat burner, an online "liver support" capsule - stop it for three months, and tell your doctor you have stopped. Then repeat your LFT. If the enzymes fall, you have your answer and you have saved money. If they do not, you have lost nothing except the cost of three months of capsules. Do not stop prescribed medicines this way; that is a different decision and needs a doctor.

And keep a list. Every product, every brand, every dose, written down and carried to appointments. The reason Indian herbal liver injury is so hard to treat is that patients cannot say what they took, and the products cannot say what is in them.

CHAPTER 19

## The medicines you already take: statins, paracetamol, TB drugs, thyroid

A common and dangerous misunderstanding is that a damaged liver means you must stop your other medicines. In most cases the opposite is true - the conditions those medicines treat are what is driving the liver damage.

| Medicine | The fear | What the guidance actually says |
|---|---|---|
| **Statins (atorvastatin, rosuvastatin and others)** | "They damage the liver, so I stopped them." | Statins are safe in fatty liver, including when transaminases are raised, and are explicitly recommended because cardiovascular disease is the leading cause of death in people with MASLD. Stopping a statin to protect the liver removes the protection your heart needed. |
| **Metformin, SGLT2 inhibitors, GLP-1 agonists** | "Diabetes medicine will strain the liver." | These are the preferred glucose-lowering options in people with MASLD. Insulin is reserved for decompensated cirrhosis. Do not change anything without your physician. |
| **Paracetamol (acetaminophen)** | "It is a liver medicine risk, so I avoid all painkillers." | Paracetamol is the safest analgesic in liver disease at correct doses - up to 3 g a day in most adults, and many hepatologists cap it at 2 g a day in cirrhosis. The danger is duplicate dosing: paracetamol hiding inside three different cold and flu combinations taken together. |
| **NSAIDs (ibuprofen, diclofenac, aceclofenac)** | "It is just a painkiller." | Avoid in cirrhosis - they cause kidney injury and bleeding. In early fatty liver they are not forbidden but should not be a daily habit. |
| **Anti-TB treatment (isoniazid, rifampicin, pyrazinamide)** | "My liver is already bad." | This combination is the leading cause of drug-induced liver injury in India, and it is also life-saving. Never stop it on your own. Instead, tell your treating doctor you have fatty liver so that baseline and follow-up liver tests are done properly. |
| **Thyroid tablets** | "Thyroid medicine affects the liver." | An untreated underactive thyroid worsens weight, cholesterol and liver fat. Treating it helps. |
| **Steroids, methotrexate, tamoxifen, amiodarone, valproate** | "I need these." | All can affect the liver. The answer is monitoring, not avoidance - a baseline LFT and periodic checks while you are on them. |

Two habits worth adopting this week. First, keep a single list of every medicine and supplement, including dose and who prescribed it, in your phone and on paper in your bag. Second, whenever a new medicine is prescribed - by any doctor, for any reason - say the words: "I have fat in my liver and my last ALT was X." That one sentence changes what gets prescribed and what gets monitored.

CHAPTER 20

# Vaccines, alcohol, and the "just one peg" question

## Vaccines

Two vaccinations matter specifically for a person with liver disease. Hepatitis A and hepatitis B are both preventable, both can cause acute liver injury, and in someone who already has liver disease either can tip them into liver failure. International guidance recommends both for people with chronic liver disease, and hepatitis B vaccination is recommended for all infants in India's universal immunisation programme. If you do not know whether you are immune, a blood test answers it - and if you are not, the vaccine series is inexpensive and widely available.

## Alcohol: the honest version

The guidance here is unusually united. The 2024 European guidelines concluded that the presumed benefits of light to moderate drinking do not hold up in people with metabolic risk factors, that any level of alcohol is associated with worsening liver outcomes in MASLD, and that alcohol should be stopped completely and permanently in anyone with advanced fibrosis or cirrhosis. The Asia-Pacific and Latin American guidelines go further and say there is no safe level at all. In a study of 8,345 people with fatty liver, alcohol intake above 10 g a day - one standard peg - doubled the risk of advanced liver disease compared with abstainers.

So the answer to "can I have just one peg" depends entirely on where you are:

| Your stage | The honest answer |
|---|---|
| **F0-F1, no fibrosis, enzymes normal** | The guidance discourages any alcohol. If you drink anyway: quantify it in grams, keep it under the MASLD threshold in Chapter 2, never on consecutive days, never on an empty stomach, and never as a binge. Tell your doctor the real number, not the polite one. |
| **F2 or F3 fibrosis** | Stop. This is the stage where the arithmetic stops being arguable - you have a measurable amount of scarring and every drink works against reversing it. |
| **Cirrhosis (F4)** | Stop completely and permanently. Even one to six drinks a week has been associated with reduced survival and more decompensation in cirrhosis, and any drinking raises liver cancer risk. |

If stopping is genuinely hard - and for a meaningful number of people it is - that is a medical problem with medical solutions, not a failure of character. De-addiction support is available through district hospitals and through the national mental health helpline (Tele-MANAS, 14416). Saying that out loud in a consulting room is the single highest-value thing a heavy drinker with fatty liver can do.

### Part Four in five sentences

Every medicine approved for this condition is approved for steatohepatitis with fibrosis - not for fat alone. If your FIB-4 is low, the treatment is Part Three.

India has two options in 2026: saroglitazar, approved here since 2020 and available nowhere else, and semaglutide 2.4 mg, approved for MASH in July 2026. Resmetirom is not yet marketed in India.

In every trial, the placebo group also improved on diet and exercise. Drugs add to lifestyle change; none replaces it.

Traditional and alternative medicines were the second largest cause of drug-induced liver injury in the Indian registry, and cases linked to them tend to be severe.

Statins are safe in fatty liver and should not be stopped. Paracetamol at correct doses is the safest painkiller. Anti-TB treatment is never stopped without your doctor.

Before you buy anything that calls itself a liver product, reread Chapter 18. It is the chapter that saves the most money and prevents the most harm.

PART FIVE

# Living with it

Monitoring, red flags, family, money, and what a year of this actually looks like.

CHAPTER 21

## Your monitoring calendar for the next five years

Fatty liver is managed on a calendar, not on symptoms - because there are no symptoms until late. This is the schedule the guidance implies, adapted for Indian costs and availability. Worksheet W7 is a blank version to fill in with your own dates.

| When | What to do | Why |
|---|---|---|
| **Today** | Weight, waist, blood pressure. Collect every old report you can find. Compute FIB-4. | You cannot measure progress from a starting point you never recorded. |
| **Month 1** | Full baseline bloods if not done: LFT, CBC with platelets, HbA1c, fasting glucose, lipid profile, TSH, HBsAg, anti-HCV. Ultrasound if the last one is over a year old. | Establishes the diagnosis properly and rules out the other causes in Chapter 8. |
| **Month 3** | Weight, waist, and a repeat ALT and GGT. No scan. | Enzymes respond within weeks if the plan is working. A scan at three months tells you almost nothing new. |
| **Month 6** | LFT, CBC, HbA1c, lipids. Recompute FIB-4. See a physician with the trend. | Six months is when the direction of travel becomes visible. |
| **Month 12** | LFT, CBC, lipids, HbA1c, ultrasound. FibroScan if you have never had one, or if FIB-4 has risen. | This is the review that tells you whether the fat has actually reduced. |
| **Every 12 months thereafter** | LFT, CBC, FIB-4, lipids, HbA1c, blood pressure, weight, waist. Ultrasound every 1 to 2 years. | Annual is enough for stable, low-risk disease. |
| **If you are F2 or F3** | Elastography every 6 months while on treatment, per the India-specific consensus on MASLD in type 2 diabetes. Specialist follow-up. | Fibrosis needs closer watching than steatosis, and response to therapy is judged on it. |
| **If you have cirrhosis** | Six-monthly ultrasound plus alpha-fetoprotein for liver cancer surveillance, endoscopy for varices as advised, and a hepatologist at least annually. | This is a different disease stage with a different surveillance protocol. It is not optional. |

One piece of advice that saves money and anxiety: do not repeat an ultrasound every three months. Fat grades move slowly, sonographers vary, and a report that says "grade 2" three months after "grade 2" is not evidence of failure - it is evidence that you did the wrong test. Track the bloods and the tape measure between scans.

CHAPTER 22

## Red flags: when to stop reading and go to a hospital

Fatty liver on its own does not cause emergencies. Cirrhosis does. If any of the following happens, the correct response is a hospital, today, not a chapter in a book.

| Symptom | What it may mean | What to do |
|---|---|---|
| **Yellow eyes or skin, dark urine, pale stools** | Jaundice - liver injury or a blocked bile duct | Same-day medical assessment. Do not wait to see whether it settles. |
| **Vomiting blood, or black tarry stools** | Bleeding from enlarged veins in the food pipe - a life-threatening complication of cirrhosis | Emergency. Call 112 or go to the nearest emergency department immediately. Do not drive yourself. |
| **Swelling of the abdomen, or swelling of both legs** | Fluid accumulation from portal hypertension | See a doctor within 24 to 48 hours. |
| **Confusion, drowsiness, unusual forgetfulness, flapping hands** | Hepatic encephalopathy | Emergency assessment. |
| **Fever with abdominal pain** | Infection of the ascitic fluid, which is dangerous in cirrhosis | Same-day hospital. |
| **Sudden severe abdominal pain** | Gallstones, pancreatitis, or a blocked vessel | Emergency. |
| **Unintentional weight loss, loss of appetite, a new lump** | Needs investigation, including for liver cancer in cirrhosis | See a doctor within a week, not a month. |
| **Easy bruising or bleeding gums with jaundice** | Reduced clotting from advanced liver disease | Same-day assessment. |

The reason this chapter exists at all is that the early stages of this disease give you nothing to feel. That is the deal fatty liver offers: twenty quiet years in exchange for doing nothing about it. The red flags above are what the end of that deal looks like. Almost nobody reading this book will reach them - most people with fat in the liver never develop cirrhosis - but the ones who do are usually the ones who never had a FibroScan.

CHAPTER 23

## Your family, your children, your kitchen

Two numbers make this chapter necessary. The pooled prevalence of fatty liver in Indian children is 35.4 per cent, and among children with obesity it is 63.4 per cent. And 93.7 per cent of adults with fatty liver in the largest Indian cohort were themselves overweight - which means this is rarely one person's problem. It is a household's cooking, a household's evening, and a household's weekend.

### The kitchen reset, in one shopping trip

- A measuring spoon for oil, kept next to the stove. Five to six teaspoons a day for the whole family is the target; the spoon is what makes it real.
- One frying pan put away, one non-stick tawa and one pressure cooker in daily use.
- No sugar-sweetened drinks in the house at all. Not reduced, not for guests. The single most effective change a family can make, because it removes the decision.
- A fruit bowl where the biscuit tin used to be, and a bowl of roasted chana or makhana on the counter.
- Whole atta, and one millet flour mixed in - jowar or bajra at 20 per cent of the atta is undetectable in a roti.
- Curd set every night. It is the cheapest protein in India and it fills the gap that snacks currently fill.

### Children: what to do and what not to do

Do not put a child on an adult weight-loss diet. Children are growing, and restriction can cost them height and muscle. The approach that works is family-level: the same food for everyone, more movement, less screen time, fixed meal times, and no commentary about the child's body at the dinner table. If a child is overweight with a large waist, or there is a strong family history of diabetes, ask a paediatrician whether an ultrasound and a liver enzyme test are warranted - the threshold for investigating children is lower than most parents assume, and the condition is reversible at that age with changes that feel like nothing.

And one thing for the adults: do not become the person who eats separately. A "diabetic plate" eaten alone at 9 pm while the family eats together is the most reliable predictor of quitting by week six. Change the family's food, not your status in it.

CHAPTER 24

## Money: tests, insurance, government schemes, and what a transplant costs

Fatty liver is one of the cheaper chronic conditions to manage well and one of the most expensive to manage late. Here is the arithmetic, so you can decide where to spend.

| Item | Typical 2026 cost | Notes |
|---|---|---|
| **Liver function test** | Rs 425 to Rs 500 at CGHS rates; Rs 300 to Rs 900 in private labs | The single most useful recurring test. Do not skip it to save money. |
| **Complete blood count (for platelets)** | Rs 255 to Rs 300 at CGHS rates | Needed for FIB-4. Often bundled with the LFT. |
| **Ultrasound whole abdomen** | Rs 680 to Rs 800 at CGHS rates; Rs 800 to Rs 2,000 private | Once a year is enough, not once a quarter. |
| **FibroScan with CAP** | Rs 978 to Rs 1,150 at CGHS rates; Rs 2,800 to Rs 7,000 private; Rs 200 to Rs 500 at government hospitals | One or two in your life, not one per quarter. |
| **HbA1c** | Rs 255 to Rs 300 at CGHS rates | Every 6 to 12 months if you have diabetes or prediabetes. |
| **Specialist consultation** | Rs 500 to Rs 2,000 private; free or nominal in government hospitals | Bring printed reports; you get more from a ten-minute consultation with paper than a thirty-minute one without. |

### What the government already provides

India was the first country to formally build fatty liver into a national NCD programme, integrating NAFLD into the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke in February 2021, and publishing operational guidelines and a training module for medical officers. That programme now runs as NP-NCD and explicitly includes fatty liver.

In practice, that means two things you can use. First, population-based screening for common NCDs - including blood pressure, diabetes and related risk factors - is available free at Ayushman Arogya Mandirs for everyone over 30; there were 173,827 of them operational as of July 2024. If you have not had a blood sugar check in a year, walk into one. Second, district and CHC-level NCD clinics can do the follow-up and referral that fatty liver needs, at low or no cost.

Ayushman Bharat PM-JAY provides Rs 5 lakh a year of secondary and tertiary hospitalisation cover for eligible families, and it covers hospitalisation, not outpatient tests. It is eligibility-based, not something you

buy.

### Insurance: the one thing you must not do

Under IRDAI's 2024 framework, the maximum waiting period for a pre-existing disease is 36 months from the start of cover, reduced from 48. Fatty liver is usually not excluded outright, but if you were diagnosed, investigated or advised treatment within 36 months before buying the policy, it falls inside the pre-existing disease framework and liver-related claims may not be paid until that waiting period is over.

So: **disclose it.** Declaring fatty liver on the proposal form costs you a waiting period. Not declaring it can cost you the entire claim years later, at the worst possible moment, because the insurer will find the ultrasound report in your file. Buy cover early, before the diagnosis, and buy enough of it - a base policy plus a super top-up is the standard way to get a large cover without a large premium.

And the number that explains why the insurance conversation matters: a liver transplant in India is commonly quoted at Rs 10 to 35 lakh at most centres, with high-end private hospitals quoted at Rs 30 to 50 lakh, and donor expenses, immunosuppressants and post-transplant care often falling outside the covered amount. That is the cost of arriving at the end of twenty untreated years. It is also, almost entirely, avoidable.

CHAPTER 25

## The honest twelve-month review: what "reversed" actually means

Twelve months in, three things can be true, and it is worth being clear about which one you are looking at.

| What happened | What it means | What to do next |
|---|---|---|
| **Ultrasound now normal, enzymes normal, waist down** | Steatosis has cleared. This is the outcome the plan is built for, and it happens in a large share of people who lose 7 to 10 per cent of body weight. | Do not go back. The metabolic tendency that caused it has not disappeared - it is being managed. Keep the three habits that got you here and re-test annually. |
| **Fat reduced but fibrosis unchanged** | The fat is responding; the scar is slower. Fibrosis regresses over years, not months, and only after sustained weight loss. | Continue. This is a normal twelve-month result, not a failure. Judge it again at 24 months. |
| **Nothing has changed** | Four honest possibilities: the plan was not followed; something unmeasured is driving it (alcohol, sleep apnoea, a medicine, an unlabelled supplement); the diagnosis is wrong (another cause in Chapter 8); or you are in the small lean-MASLD group where weight is not the main lever. | Do not escalate to buying supplements. Go back to Chapter 9, rebuild the risk map with fresh numbers, and take it to a physician with a specific question. |

A word about the word "reversed", because it is used loosely by everyone selling something. Fat in the liver can genuinely clear - that is not marketing, it is histology. Fibrosis can regress, more slowly and less completely. What does not happen is a cure: the insulin resistance, the visceral fat tendency and the genetic hand you were dealt remain, and if the food and movement patterns go back to what they were, the fat comes back. What you are buying with these twelve weeks is not a finished state. It is a different trajectory.

> **The five-year view**

> If you hold roughly this pattern for five years - waist under the cut-off, 7 per cent of body weight off and kept off, no sugary drinks, 200 minutes a week of movement, no more than the occasional drink, and an annual blood test - the probability that fatty liver ever becomes the thing that puts you in a hospital is small. That is the entire promise of this book, and it is a modest, honest one. Nothing here is faster than that, and anything that claims to be is selling you something.

### Part Five in five sentences

This is managed on a calendar, not on symptoms: bloods at three and six months, a full review at twelve, then annually - and elastography every six months if you are F2 or F3.

Learn the red flags. Jaundice, vomiting blood, a swelling abdomen or new confusion mean a hospital today, not a chapter in a book.

Change the household's food, not just your own plate, and do not put a child on an adult diet - take the family to a paediatrician instead.

Free screening for over-30s is available at Ayushman Arogya Mandirs, and India's NCD programme formally includes fatty liver. Use it before you pay for anything.

Disclose the diagnosis when you buy health insurance. The waiting period is capped at 36 months; a non-disclosure can cost you the entire claim.

The last chapter is the one to reread at month twelve, when the novelty has gone and the numbers are the only thing left to argue with.

PART SIX

# Worksheets

Eight pages to print, fill in and take to your next appointment.

## W1 MY REPORT CARD

Fill this in once, then re-date it every time you get new numbers. The point is the column of dates, not the first row.

| What | Baseline (date) | 6 months | 12 months | Target |
|---|---|---|---|---|
| **Weight (kg)** | | | | Down 7 per cent |
| **Waist (cm)** | | | | Under 90 men / 80 women, or half your height |
| **BMI (Indian cut-offs)** | | | | Under 23 |
| **ALT (U/L)** | | | | Falling towards normal |
| **AST (U/L)** | | | | Lower than ALT |
| **GGT (U/L)** | | | | Falling |
| **Platelets** | | | | Stable or rising |
| **FIB-4** | | | | Under 1.3, ideally under 1.0 |
| **HbA1c (per cent)** | | | | Under 5.7, or under 7 if diabetic |
| **Triglycerides** | | | | Under 150 |
| **HDL** | | | | Above 40 men / 50 women |
| **Blood pressure** | | | | Under 130/80 |
| **Ultrasound grade / CAP** | | | | Lower than baseline |
| **LSM (kPa)** | | | | Under 8.2, ideally under 6 |
| **Alcohol (g per day)** | | | | 0, or under the MASLD threshold |

## W2 FIB-4 CALCULATOR SHEET

Work through it in the order written. Use the numbers exactly as printed on your report.

| Step | What to write |
|---|---|
| **1. Your age in years** | |
| **2. AST, as printed (U/L)** | |
| **3. Multiply step 1 by step 2** | |
| **4. ALT, as printed (U/L)** | |
| **5. Square root of step 4 (use the calculator's square-root key)** | |
| **6. Platelet count as a plain number, e.g. 245 for 2.45 lakh** | |

| Step | What to write |
|---|---|
| **7. Multiply step 5 by step 6** | |
| **8. Divide step 3 by step 7 - this is your FIB-4** | |
| **9. Which band: below 1.0 / 1.0-1.3 / 1.3-2.67 / above 2.67** | |
| **10. Your action from Chapter 6** | |

Cautions to note on this sheet: FIB-4 reads low under about 35 years of age and high over 65; platelets fall for reasons unrelated to the liver (dengue, some medicines); do not compute it during an acute illness. It estimates risk - it does not measure fibrosis.

## W3 TWELVE-WEEK WEIGHT AND WAIST LOG

Weigh yourself once a week, same time, same clothes. Measure the waist once a fortnight.

| Week | Weight | Waist | What I changed this week | Energy (1-5) |
|---|---|---|---|---|
| **1** | | | | |
| **2** | | | | |
| **3** | | | | |
| **4** | | | | |
| **5** | | | | |
| **6** | | | | |
| **7** | | | | |
| **8** | | | | |
| **9** | | | | |
| **10** | | | | |
| **11** | | | | |
| **12** | | | | |

## W4 WEEKLY PLATE AUDIT

Do this on a Sunday for the week just gone. Count the days, not the intentions.

| Question | Days this week (0-7) | What I will change |
|---|---|---|
| **Did I have a sugar-sweetened drink or packaged juice?** | | |
| **Did I eat something deep-fried?** | | |
| **Did I have protein at breakfast?** | | |
| **Was half my lunch or dinner plate vegetables?** | | |
| **Did I eat within three hours of going to bed?** | | |
| **Did I keep total oil to about 5 to 6 teaspoons?** | | |
| **Did I eat fruit instead of a sweet or biscuit?** | | |

## W5 MOVEMENT LOG

Tick what you actually did. Twenty honest minutes beat a planned hour you skipped.

| Week | Aerobic minutes | Strength sessions | Post-meal walks | Notes |
|---|---|---|---|---|
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |
| 7 | | | | |
| 8 | | | | |
| 9 | | | | |
| 10 | | | | |
| 11 | | | | |
| 12 | | | | |

## W6 MEDICINE AND SUPPLEMENT LIST

Fill this in today and keep a copy in your bag. Take it to every appointment.

| Medicine or product | Dose | Who prescribed it | Why I take it | Start date |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |

## W7 TEST CALENDAR

Book the next two dates now, in your phone. Everything after that follows the pattern in Chapter 21.

| Test | Last done | Next due | Where | Approx cost |
|---|---|---|---|---|
| LFT | | | | |
| CBC with platelets | | | | |
| HbA1c | | | | |
| Lipid profile | | | | |

| Test | Last done | Next due | Where | Approx cost |
|---|---|---|---|---|
| **Ultrasound abdomen** | | | | |
| **FibroScan** | | | | |
| **Blood pressure check** | | | | |
| **Weight and waist** | | | | |

## W8 TEN QUESTIONS FOR YOUR DOCTOR

Circle the ones that apply to you. Take the answers in writing - a photograph of the page is fine.

1 What is my fibrosis stage, and how do you know? Have I had a FIB-4 or a FibroScan?
2 Is my ALT normal, and does that mean my liver is fine?
3 Do I meet the criteria for MASLD, and which of the five do I meet?
4 Should I be checked for hepatitis B, hepatitis C, thyroid disease or coeliac disease?
5 Which of my current medicines or supplements could be affecting my liver?
6 Am I a candidate for a liver-directed medicine, or is lifestyle change the right treatment for now?
7 What is my target weight and waist, and over what period?
8 Do I need a sleep study?
9 When should I repeat the blood tests, and what result would make you change the plan?
10 At what point should I see a hepatologist rather than a physician?